CHAPTER 1

INTRODUCTION

1.1 Background:

Autism spectrum disorder (ASD) is characterized by a typical development affecting communication, behavior, and social functioning, among which the degree of severity varies greatly. The Centers for Disease Control and Prevention (CDC's) Autism and Developmental Disabilities Monitoring (ADDM) Network reports 1 in 31 (3.2%) 8-year-olds in the U.S. have ASD. This makes the neurodevelopmental disorder one of the most prevalent conditions in the international community, thus early intervention and tracking of behavioral changes become critical to improve outcomes and enhance quality of life.

Though public knowledge of ASD is rising significantly, the pathway from raising concerns to diagnosis tends to be prolonged, expensive, and confined to medical institutions. Standard methods of diagnosing autism include a thorough behavioral assessment, the use of screening tests such as the Autism Diagnostic Observation Schedule, and expertise of a specialized clinician, the availability of which remains uneven across different contexts and socio-economic levels. Families living in underserved or faraway areas have to endure long periods of waiting before having a diagnosis, while losing valuable time for intervention.

1.2 Problem Statement:

The current approaches to monitoring and evaluating ASD have various weaknesses:

1) The dependency on the clinic and specialists ensures that any diagnosis and treatment depend on qualified people and require appointments, making it inaccessible for families living far away from such places.

2) Security concerns arise because children diagnosed with ASD may pose a danger to themselves, which will not be addressed in the absence of their caregivers.

3) It is based on subjective assessment; behaviors are evaluated based on an individual's interpretation, thus causing differences between evaluators and within different assessment periods.

4) The evaluation process is continuous, and regular day-to-day behaviors cannot be captured since

they are done in one's natural environment, such as the home.

5) The cost incurred is quite high due to frequent appointments with specialists and treatments.

1.3 Objectives:

1) To develop a robotic companion for kids with ASD to detect self-harm to trigger immediate sound diversion in real time to protect the child.

2) To observe the child’s actions, monitor their ASD behavior, and engagement to generate structured reports for parents and healthcare professionals automatically.

3) To feature an AI-enabled conversation tool that would make the interaction with the child more human. An interactive learning toolkit will teach kids alphabets, numbers, shapes, colors, and words using voice recognition features. Parents will receive a web dashboard to monitor the process via live camera feed and remotely manage the robot from anywhere.

CHAPTER 2

LITERATURE SURVEY

2.1 Social Robots for ASD Therapy

There have been numerous applications of the use of robots in ASD intervention strategies recently due to technological advancements in robotics, artificial intelligence, and machine learning algorithms. In particular, a meta-analysis involving an examination of 107 articles written between 2005 and 2025 concerning the topic of robot-assisted ASD intervention, published in MDPI's Multimodal Technologies and Interaction, pointed out that social engagement, autonomous functioning, and adaptation were among the essential aspects of successful robotic interventions in ASD treatment, while there was also observed a considerable increase in deployment studies beginning in 2019 by Muhammad et al. [1]. Furthermore, in a randomized controlled trial, by Vasiliki et al. [4] the application of a humanoid robot-assisted psychosocial treatment program for ASD affected children demonstrated considerable benefits in terms of psychosocial performance as indicated through neuropsychological tests and parents' observation; thus, social robots may indeed be considered a valuable addition to traditional therapy. Moreover, Kouroupa et al. [11] provided a systematic review and meta-analysis of studies on the effectiveness of social robots' application within the ASD population, finding evidence for improvement of the mentioned outcomes while emphasizing the importance of personalization and adaptability.

2.2 Clinical data and electronic health

With the increasing availability of big clinical records stored electronically, a novel breed of prediction algorithms has emerged, which uses data gathered sequentially from medical histories. The work done by Grosvenor et al. [3] proved the possibility of integrating machine learning algorithms into the current health information infrastructure, thereby enabling early identification of children who are at risk up to several months ahead of clinical evaluations by using data from routine clinical visits. Ben-Sasson et al. [2] developed an algorithm that predicts autism diagnosis based on well-baby electronic health records, providing clinically useful predictions.

2.3 Machine Learning Approaches for ASD Detection

Several studies have been conducted to explore the potential of using supervised machine learning algorithms on structured data for detecting ASD. Meng et al. [7] presented a machine learning model based on eye tracking as a means of diagnosing ASD with high accuracy, which essentially provided biometric data to be added to the emerging range of automatic methods of ASD diagnosis. Akter et al. [21] found that ensemble models can successfully be used for early-stage diagnosis of multi-age patients with ASD by processing information contained in different age groups, ranging from toddlers to adult patients. Parikh et al. [20] enhanced the performance of ML pipelines through the use of personal characteristics of subjects, proving that thoughtfully selected features can significantly increase the accuracy of diagnosis despite limitations in the amount of training data. Thabtah and Peebles [19] presented an interpretable rule-based machine learning model for autism diagnosis among diverse demographic profiles. This research highlighted the importance of clinical interpretability in ML models for autism diagnosis since they were used in varied demographic populations. Extending from their work, Thabtah et al. [22] proposed a computational intelligence model which provided simple decision rules. Raj and Masood [12] evaluated various classifiers on standard datasets of ASD to show that gradient boosting and random forest were always superior to basic classifiers in terms of sensitivity and specificity. On the other hand, Hossain et al. [13] did a longitudinal study using ASD datasets collected in adults and children over eleven years. This helped to develop detection methods based on benchmarks of various classifiers across different demographics and to provide guidelines for developing ASD classifiers. Vakadkar et al. [14] discussed various ML models to classify autism in children without needing special equipment.

2.4 Computer Vision and Behavioural Recognition

Alongside other work, studies focused on the use of computer vision in the passive surveillance of the behavioral characteristics of autistic children through non-invasive means. Ganjigunte Prakash et al. [9] provided an in-depth discussion of an automated assessment pipeline comprising analysis of social behaviors, emotion detection, body postures, and activities involving the practical skills of children with autism. This demonstrates the benefit of incorporating multiple visual signals into a single framework in behavioral assessments. Also, Wei et al. [8] conducted a comprehensive study on activity recognition systems based on vision in children exhibiting behaviors associated with autism. The results revealed that CNN was able to accurately identify different types of activities associated with ASD from video input data. Negin et al. [12] used vision-assisted techniques in detecting repetitive ASD behavioral patterns. Using camera-based observations, they found it possible to accurately identify stereotypical movements such as arm flapping and rocking without relying on any wearable devices.

2.5 Pose estimation for ASD analysis

In the same vein, pose estimation techniques have been shown to be effective non-invasive tools in phenotyping behavioral traits in ASD. In a systematic review by Simeoli et al. [5] evaluated in detecting early signs of ASD using motion data. According to their findings, features extracted using poses have been found to exhibit consistency across multiple populations of pediatrics, particularly features obtained using OpenPose. Kojovic et al. [15] used OpenPose features of 2D skeletal key points of ADOS videos that were later fed into CNN-LSTM models using non-verbal behaviors exclusively. Such an innovative technique successfully discriminated between ASD children and typically developed children at a high level of accuracy, indicating pose estimation as a promising marker-free diagnostic modality. The establishment of such an approach can be attributed to the groundbreaking work by Cao et al. [16] with the development of OpenPose, which is a multi-person real-time system for estimating 2D pose using part affinity fields.

2.6 Self Injurious Behaviour Detection

Self-harm and self-stimulating behaviors among children with autism have been considered due to the direct risks involved. For instance, in a study by Cantin-Garside et al. [17] machine learning algorithms were employed to classify self-harming behaviors in people diagnosed with autism, using sensory input and video information. It was made clear just how difficult differentiating between such behavior and regular play motions could be; thus, the need for accurate classification systems. In another set of research, Rajagopalan et al. [23] investigated the classification of self-stimulating behaviors in situations where physical constraints were absent through the use of motion history images.

CHAPTER 3

PROPOSED SOLUTION

3.1 Overview

The proposed system is a solution for AI robots used in the home environment of children diagnosed with Autism Spectrum Disorder. It combines features such as behavior classification, robotic therapies, safety monitoring, interactive learning, and the caregiver clinical dashboard to provide a better interactive and a safety environment to the ASD patients.

The technology is based on two tightly interconnected layers. The physical hardware layer allows the robot to perceive its surroundings visually and physically, while the software layer allows it to process information from the sensors and act accordingly, communicating with the caregiver. Altogether, this architecture gives the system the ability to be an intelligent, emotionally expressive robot companion that is constantly observing the child, ensuring safety when needed, facilitating learning activities, and monitoring behavior objectively.

3.2 Operational Modes

As shown in the flowchart below, the system operates in four modes. On startup, the system defaults to Idle Mode. An event trigger — either an automatic detection or a caregiver command — switches the system into one of the three active modes. Upon completion, all modes return to Idle. All mode activity is logged to Firebase regardless of which mode is active.

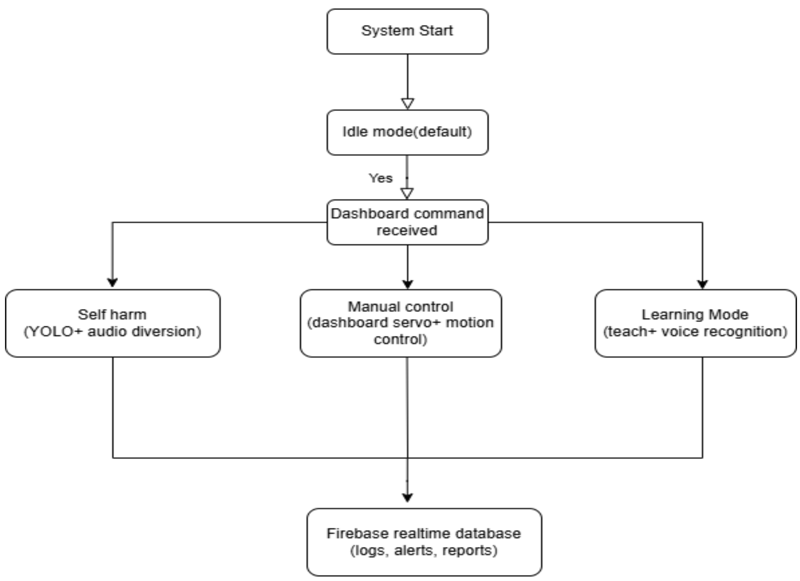

Fig. 3.2 Flowchart of operational Mode

3.3 Data flow and reporting

All data obtained from each session including the behavioral scores obtained via OpenPose, self-harm incidence reports, performance in learning sessions, and robot's state data is collected in real-time into Firebase Real-time Database. The dashboard for caregivers provides all this data compiled into informative reports, which include trends in engagement, reported behavior incidences, and overall learning progression of patients. This way, clinicians get a source of continuous behavioral data to supplement their clinical observations.

3.4 System advantages

1) The proposed solution operates continuously throughout the day rather than during periodic clinical check-ups. Continuous monitoring of the behavioral pattern of a child is done according to the course of daily activity performed by the child.

2) Immediate protection mechanism. The self-harm detection algorithm implemented in YOLO immediately triggers redirection of the audio signal, thus acting much quicker than any caregiver that may not be in the immediate vicinity.

3) Non-invasive nature. Wearable devices are not required since only a camera and a microphone are utilized in the proposed solution. Thus, children with ASD do not have any invasive elements installed on them.

4) Affordable design. The hardware components are easily accessible and inexpensive (e.g., Raspberry Pi 5, PCA9685, L298N, LiPo batteries).

5) A real-time monitoring dashboard allows parents or clinicians to have objective information about the current status and behavior of the child without scheduling a visit.

CHAPTER 4

SYSTEM ARCHITECTURE

4.1 System Overview

The proposed system operates using two interrelated layers. One is the hardware layer that consists of the components responsible for sensing, acting, and power. The other layer is the intelligence software layer that processes information, makes decisions, and communicates remotely. The two layers are closely tied together by means of the Raspberry Pi 5 that acts as an interface between the robot hardware and the cloud computing systems. The upper diagram depicts the system block diagram with all the components connected through the Pi. Fig. 4.1 shows the flowchart of the proposed system.

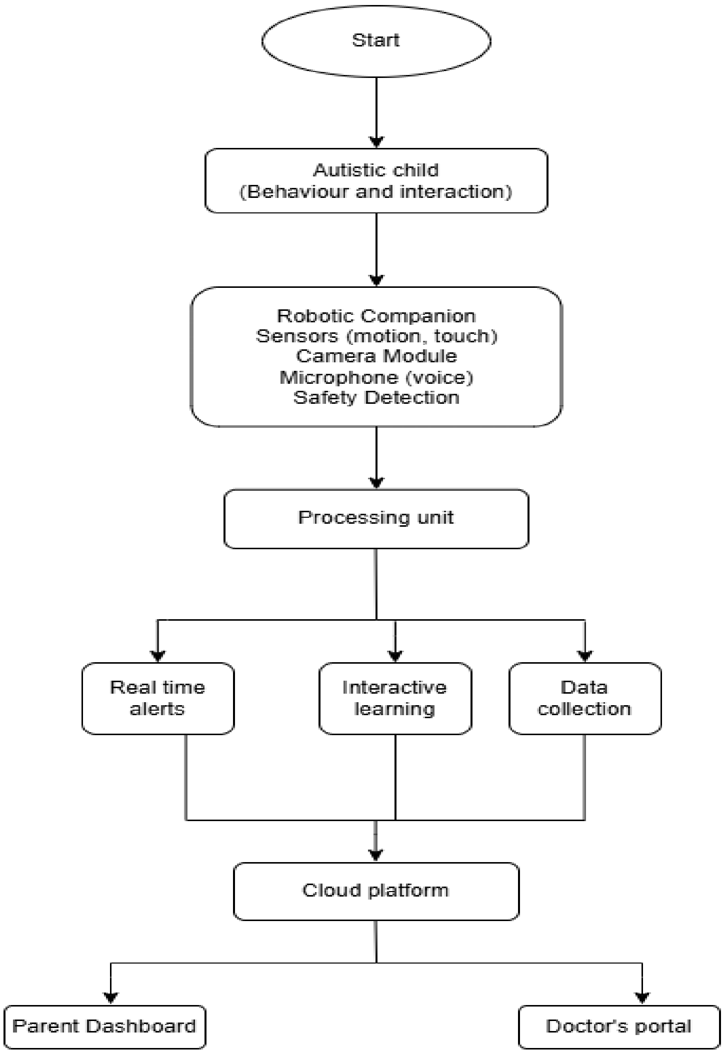

Fig. 4.1 Flowchart of the proposed system

4.2 Hardware design

The hardware model is further classified into four distinct parts namely- power subsystem, control subsystem, expression system, locomotion system.

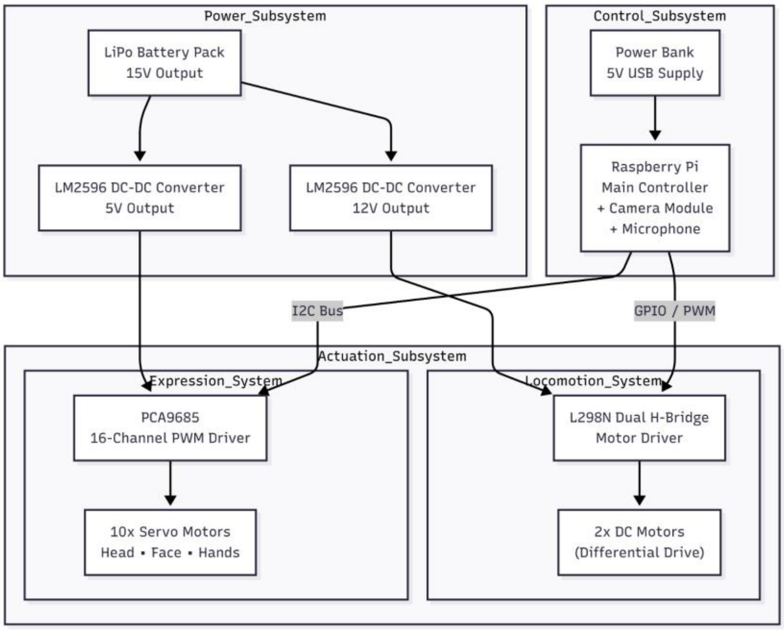

Fig. 4.2.1 Flowchart of the hardware model

a) Power Subsystem- The Power Subsystem is responsible for supplying stable and regulated voltage to all hardware components of the robot. Since the robot integrates high-current actuators (DC motors), precision devices (servos), and a sensitive computing unit (Raspberry Pi), careful power separation and voltage regulation are essential to ensure stable operation. The main components of power subsystem are:
   1. 15V LiPo battery pack- the main energy source, responsible for supplying high current.
   2. LM2596 Buck Converter (12V)- It powers L298N Motor Driver, supporting DC motor movement.
   3. LM2596 Buck Converter (5V)- powers PCA9685 Servo Driver
   4. Power Bank (5V USB)- Independently powers Raspberry Pi

b) Processing Subsystem- The Processing Subsystem acts as the central computational unit of the robot. Raspberry Pi is used for the same. It is responsible for:
   1. Speech Training
   2. Engagement scoring
   3. Open Pose integration
   4. API Calling

c) Perception Subsystem- The Perception Subsystem enables the robot to sense and interpret its surrounding environment. It collects multimodal input (audio and visual data) and converts raw signals into structured information for higher-level processing. It consists of-

1. Microphone
2. Camera
3. Speaker Output

d) Actuation Subsystem- The Actuation Subsystem is responsible for executing physical responses generated by the software system. It translates digital control signals into mechanical motion and expressive gestures. It consists of:

1. Locomotion System- It consists of L298N motor driver, that controls two DC motors using H-bridge circuitry.
2. Gesture System- It consists of PCA9685 motor driver control 10 servo motors.

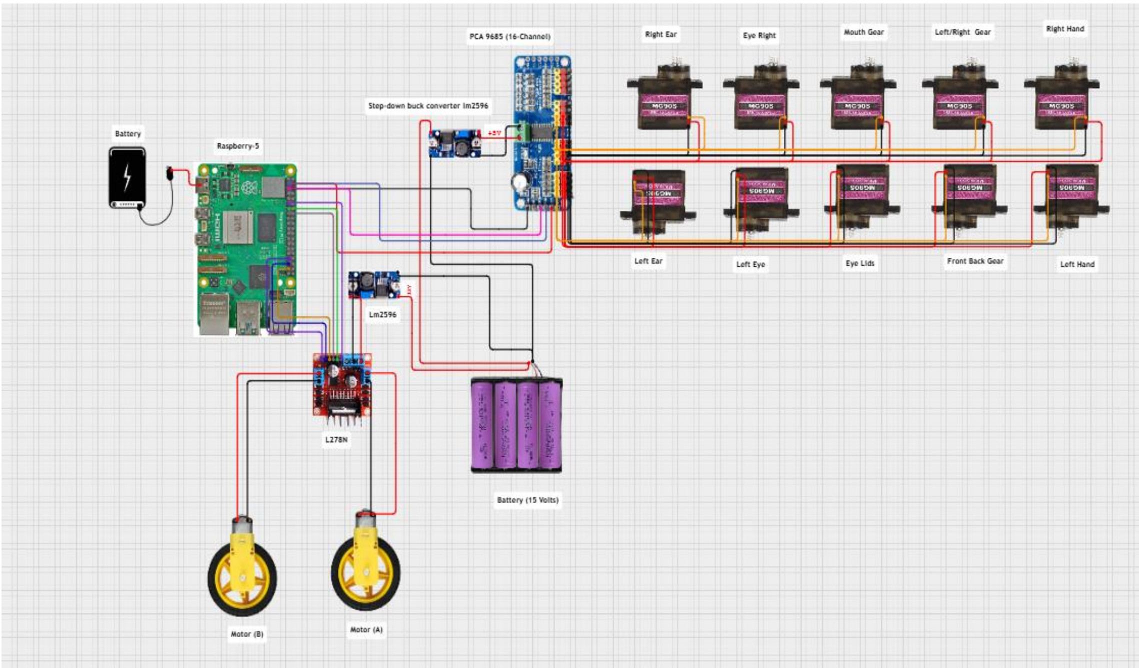

Fig 4.2.2 Circuit diagram of hardware connections

4.3 Software Design

4.3.1 Thread architecture- aria_main.py

The robot's master controller aria_main.py runs as a single Python process on the Raspberry Pi 5, managing all subsystems concurrently through a multi-threaded architecture.

Table 4.3.1 The multithread architecture

Thread	Role
Main thread	Open cv preview window

Capture_thread	Reads frames from webcam continuously at 20fps
Detection_thread	YOLO inference and ASD/self-harm analysis on alternate frames
Encode_thread	JPEG encoding of processed frames for MJPEG stream
Flash_thread	Flask HTTP server on port 5000 serving the MJPEG stream
Ngrok_thread	Starts ngrok tunnel and publishes public URL to Firebase
Heartbeat_thread	Firebase keepalive ping every 15 seconds
Idle_anim_thread	Triggers a random animation every 10 seconds in Idle Mode
Asd_push_thread	Pushes rolling ASD behavioral metrics to Firebase every 2 seconds
Voice_cache_thread	Pre-generates all gTTS audio files at startup in background
Mic_init_thread	Detects and calibrates USB microphone for ambient noise
Firebase_listener	Persistent stream listener on /commands, fires on dashboard input

4.3.2 Operational Modes

The system operates in namely 3 modes- Learning mode, Open pose, Normal Interaction mode. On starting the system automatically starts with the learning mode and then switches according to the given command.

Table 4.3.2 Operational modes of the system

Mode	Trigger	Robot Behaviour	Camera	ASD Tracking
Idle	Default on startup	Random animations every 10s with matching voice	YOLO pose tracking active	Full tracking pushed to Firebase every 2s
Self-Harm Detection	Dashboard switch or Pi M key	YOLO switches to harm detection, calming animations on detection	YOLO harm detection active	Paused
Manual Control	Dashboard switch	Responds only to slider and button commands from dashboard	YOLO continues running	Paused

Learning	Dashboard switch	Speaks learning content, plays animations, listens via mic	YOLO pose tracking active	Continues running

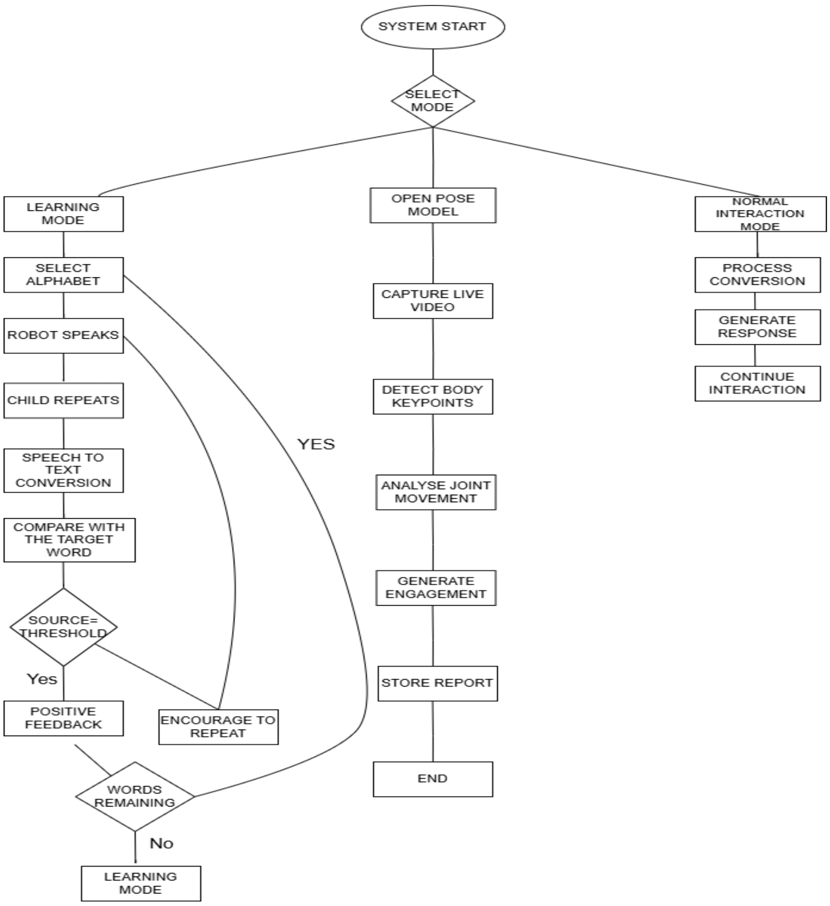

Fig 4.3.2 Detailed flowchart of the operational modes of the system

4.3.3 ASD behavioural Metrices

In Idle Mode, continuous tracking of five behavioral indicators are carried out that are derived from YOLOv8 pose keypoints. Metrics are calculated as rolling averages of the last 30 samples and pushed to Firebase every 2 seconds for live dashboard display. These metrics are also compiled into the downloadable clinical report.

Table 4.3.3 Depicts the behavioural metrices of the system

Metric	**Keypoints Used**	**Calculation Method**	**Clinical Relevance**
Body Symmetry	Shoulders (L/R), Hips (L/R)	100 minus absolute Y-difference of paired landmarks multiplied by 200, averaged	Body asymmetry can indicate motor development concerns

Posture Score	Mid-shoulder, Mid-hip	Spine angle via arctan2, deviation from 90° subtracted from 100	Posture irregularities are commonly observed in ASD
Engagement Score	Nose position, wrist presence	60 base + 25 for eye contact + 15 for calm head movement	Engagement correlates with attention and social responsiveness
Wrist Velocity	Left wrist, Right wrist	Mean velocity of position deque over 20 frames	Elevated wrist velocity indicates hand-flapping or self-stimulation
Hand Flapping	Both wrists	Combined wrist velocity exceeds 0.03 threshold per frame	Hand flapping is a stereotyped motor behavior characteristic of ASD

4.3.4 Behavioural Schema

Firebase Realtime Database serves as the central message bus between the Raspberry Pi and the web dashboard. All data is stored as JSON and synchronized in real time to all connected clients. The database is organized into six primary nodes:

Table 4.3.4 Behavioural nodes of the system

Command	Work
/commands	Dashboard Writes/ PI reads
/ status	PI writes/ dashboard reads
/ asd	PI writes every 2sec, dashboard reads
/alerts	PI pushes on self harm detection, dashboard reads.
/ stream	Pi writes on tunnel start, Dashboard reads
/ learn_result	Pi writes after mic listen, Dashboard reads

4.3.5 Web Dashboard

Caregiver Dashboard is an app made with React 18 and Vite technologies, hosted by Vercel. The app uses tabbed navigation with five pages, and everything is placed in a fixed viewport, designed specifically for tablets and phones within the context of the robot station.

4.3.6 Algorithms and model design

1. Two-Stage Hybrid Classification:

To balance computational efficiency and classification accuracy, the system employs a two-stage hybrid detection framework.

Stage 1: Rule-Based Optical Flow Screening- A fast deterministic approach is used to classify high-confidence cases directly based on motion characteristics:

- High motion intensity with dominant vertical flow → **Hard SIB**
- Dominant horizontal motion → **Stimming**
- Low overall motion → **Safe behaviour**

This stage operates in real time with minimal computational overhead and filters out easily identifiable cases, reducing the workload for subsequent processing.

Stage 2: Machine Learning Classification: Frames that are not confidently classified in Stage 1 are passed to a stacked ensemble learning model for deeper analysis.

Support Vector Machine (SVM): The SVM classifier determines an optimal hyperplane that separates different behavioural classes by maximizing the margin between them.

$$f(x) = w^T x + b \tag{i}$$

Random Forest: Random Forest improves robustness by aggregating predictions from multiple decision trees, reducing overfitting and improving generalization.

$$\hat{y} = \frac{1}{N}\sum_{i=1}^{N} T_i\,(x) \tag{ii}$$

Gradient Boosting: Gradient Boosting enhances prediction accuracy by sequentially correcting errors made by previous models.

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x \tag{iii}$$

Stacked Ensemble Learning: The outputs of all classifiers are combined using a meta-classifier to generate the final prediction. This approach leverages the strengths of individual models, resulting in improved classification performance.

2**. Pose Estimation Algorithm**

Pose estimation is used to extract human skeletal keypoints from video frames. Each frame is represented as a set of $n$body joints:

$$P = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\} \tag{iv}$$

where (x_i, y_i)represents the coordinates of the i^{th}keypoint.

Distance Between Keypoints: To understand body posture and movement, the system calculates the distance between two body joints (keypoints), such as the distance between the hand and the head or between both hands. It is given by:

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \tag{v}$$

where:

- (x_i, y_i)= coordinates of keypoint i
- (x_j, y_j)= coordinates of keypoint j
- d_{ij}= Euclidean distance between the two keypoints

Motion Velocity: To detect movement intensity and repetitive actions, the system calculates the velocity of each keypoint across consecutive frame. It is given by:

$$v_i = \frac{\sqrt{(x_i^t - x_i^{t-1})^2 + (y_i^t - y_i^{t-1})^2}}{\Delta t} \tag{vi}$$

where:

(x_i^{t-1}, y_i^{t-1})= position in previous frame

Δt= time difference between frames

v_i= velocity of keypoint i

Application:

- Detect repetitive movements (e.g., hand flapping)
- Identify abnormal posture patterns

3. Eye Contact Detection Algorithm

Eye contact is estimated using gaze direction and facial orientation extracted from detected facial landmarks.

Gaze Angle Calculation: This computes the angle of gaze relative to the face. It helps determine whether the person is looking forward (making eye contact) or away.

$$\theta = \tan^{-1}\left(\frac{y_{eye} - y_{center}}{x_{eye} - x_{center}}\right) \tag{vii}$$

where:

- (x_{eye}, y_{eye})= eye position
- (x_{center}, y_{center})= face center

4. Behavioral Pattern Analysis

The system analyzes behavior using temporal metrics such as frequency and duration.

Repetitive Behavior Frequency: This calculates how often a particular behavior occurs within a time frame.

$$f = \frac{N_{actions}}{T} \tag{viii}$$

where:

- $N_{actions}$= number of repeated actions
- T= observation time

5. Self-Harm Detection Algorithm

Self-harm detection is based on combining motion intensity, spatial proximity, and advanced motion analysis techniques such as optical flow and machine learning classification. This enables accurate identification of Self-Injurious Behaviour (SIB) in real time.

Motion Intensity

$$I = \sum_{i=1}^{n} v_i \tag{ix}$$

where:

- v_i= velocity of each keypoint
- n= number of keypoints

Directional Flow interpretation- Directional decomposition enables the system to distinguish between harmful behaviours and non-harmful repetitive actions with higher accuracy. The motion vectors that are obtained are decomposed into horizontal (f_x) and vertical (f_y) components to analyze the direction of movement.

- $f_x \ll f_y$-Indicates dominant vertical motion, typically associated with **head-banging**, which is categorized as **Hard Self-Injurious Behaviour (SIB)**.
- $f_x \gg f_y$ -Indicates dominant horizontal motion, commonly observed in **spinning behaviours**, classified as **Stimming**.
- $f_x \approx 0$ and $f_y \approx 0$- Represents negligible movement, corresponding to **Safe or idle behaviour**.

Temporal (Sliding Window) Analysis**-** To ensure reliable detection of behavioural patterns over time, the input video stream is segmented into overlapping temporal windows. The use of overlapping windows ensures continuity in analysis and prevents short-duration events, such as sudden head impacts, from being missed

- Window Size (W) = 30 frames
- Stride (S) = 8 frames

Each temporal window is classified into one of the following categories:

- Safe – Normal behaviour
- Stimming – Repetitive but non-harmful behaviour
- Soft SIB – Mild or potentially harmful behaviour
- Hard SIB – Severe self-injurious behaviour

If Hard SIB is detected:

- An immediate alert is triggered
- The system activates an audio diversion mechanism to interrupt the behaviour
- A notification is sent to the caregiver via the web dashboard

Motion Feature Extraction

From each temporal window, a set of statistical and dynamic features are extracted to characterize behavioural patterns:

- Flow Acceleration: Represents the rate of change of motion intensity over time. Sudden spikes indicate abrupt movements such as impact events.
- Motion Entropy:

$$H = -\sum p(i) \log(p(i) \quad \text{(x)}$$

where:

$p(i)$represents the probability distribution of motion magnitudes.

Interpretation:

- High entropy → chaotic and irregular motion (indicative of SIB)
- Low entropy → rhythmic or stable motion (stimming or safe behaviour)

Escalation Rate: Measures the increase in motion intensity across the window. Positive values indicate **escalating behaviour**, which is often associated with self-injurious actions.

$$E = M_{last} - M_{first} \quad \text{(xi)}$$

Flow Variance Trend: Increasing variance suggests instability and unpredictability in movement patterns.

$$\sigma^2 = \frac{1}{N} \sum (M_i - \mu)^2 \quad \text{(xii)}$$

where $\mu$is the mean motion magnitude.

4.3.7 Deployment

The system is deployed across three platforms — the Raspberry Pi 5 for the robot layer, Vercel for the React dashboard, and Firebase Hosting for the HTML fallback.

CHAPTER 5

RESULTS AND DISCUSSION

The suggested two-stage hybrid SIB classifier was tested using Hybrid Classifier-confusion matrix, Precision recall curves, Feature importance-Random forest, 5 fold cross-validation. The two-stage architecture also improves efficiency by allowing rapid rule-based screening in Stage 1, while Stage 2 performs detailed machine-learning-based classification for ambiguous cases. Table 5.1 depicts the final test performance of the hybrid model and Table 5.2 presents a comparison of the existing methods in terms of accuracy F1 score and recall.

Table 5.1 The the final test performance of the hybrid model.

Metric	Test Set	CV Mean & STD
Overall accuracy	91.5%	91.5% ± 0.8%
Hard SIB recall	93.1%	93.1% ± 1.2%
Hard SIB precision	72.8%	72.8% ± 1.5%
Hard SIB F1-Score	78.9%	78.9% ± 1.3%
Hard SIB Support	5.2%	5.2% ± 0.4%
Soft SIB Recall	79.2%	78.8% ± 1.4%
Stimming Recall	85.7%	85.3% ± 1.1%

Table 5.2 Comparison of the existing methods in terms of accuracy F1 score and recall

Author	Accuracy	F1-score	Recall
Proposed Method	**91.5**	**93**	**91**
Hossain et al. [13]	89	88	87
Vakadar et al. [14]	87	86	85
Akter et al. [21]	87	85	84
Raj and Masood [18]	88	86	85
Erkan & Thanh et al. [22]	87	86	85

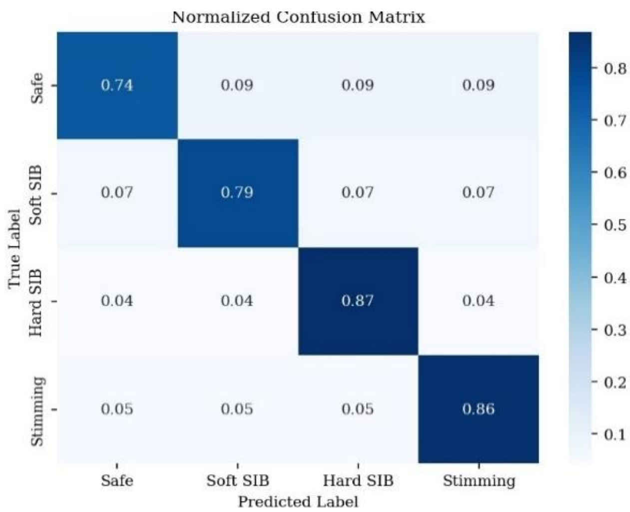

Fig. 5.1 Confusion Matrix of the hybrid classifier

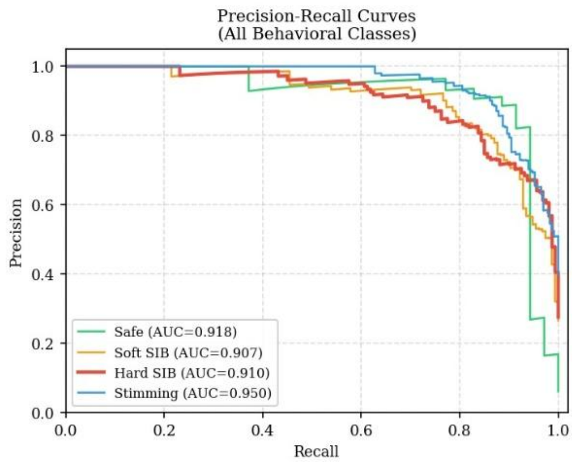

Fig. 5.2 Precision Recall curves

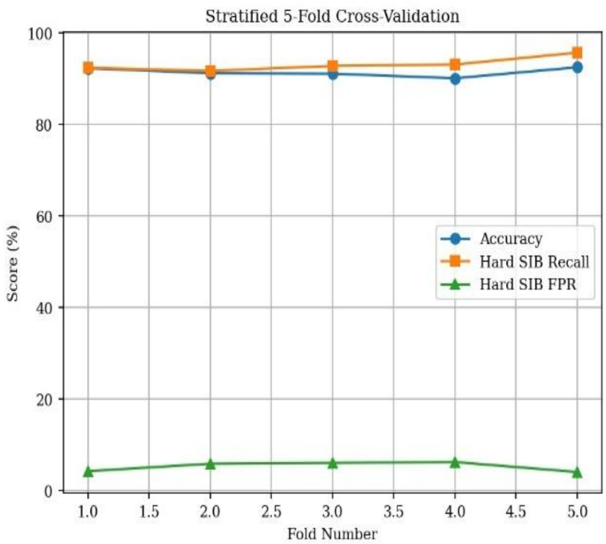

Fig 5.3 5 fold cross validation

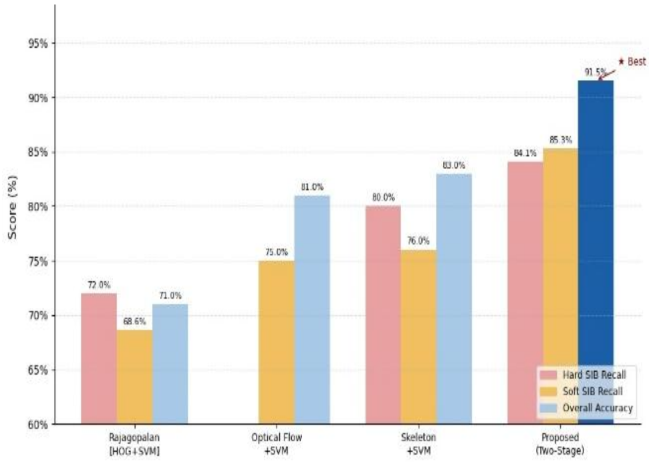

Fig 5.4 Comparison of the proposed methods with other state of art methods

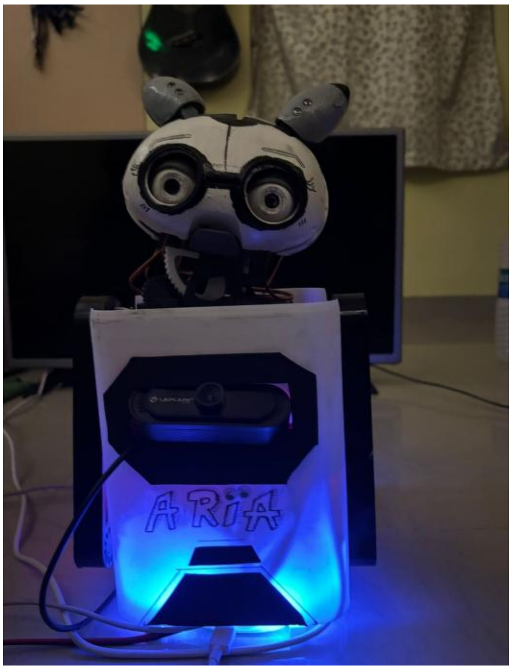

Fig 5.5 Final Robot prototype (front view)

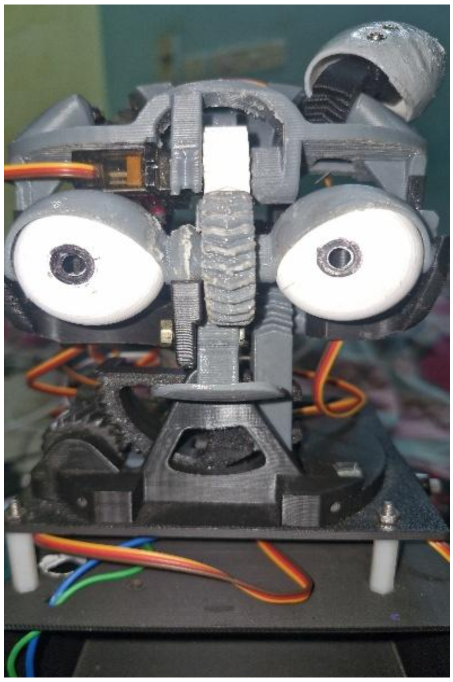

Fig 5.6 Head mechanism assembly

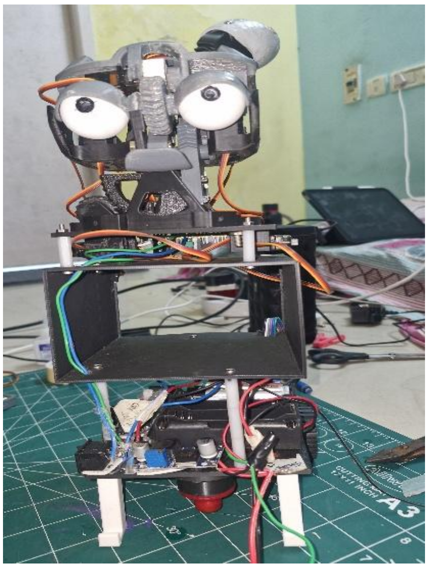

Fig 5.7 Internal hardware structure of the prototype

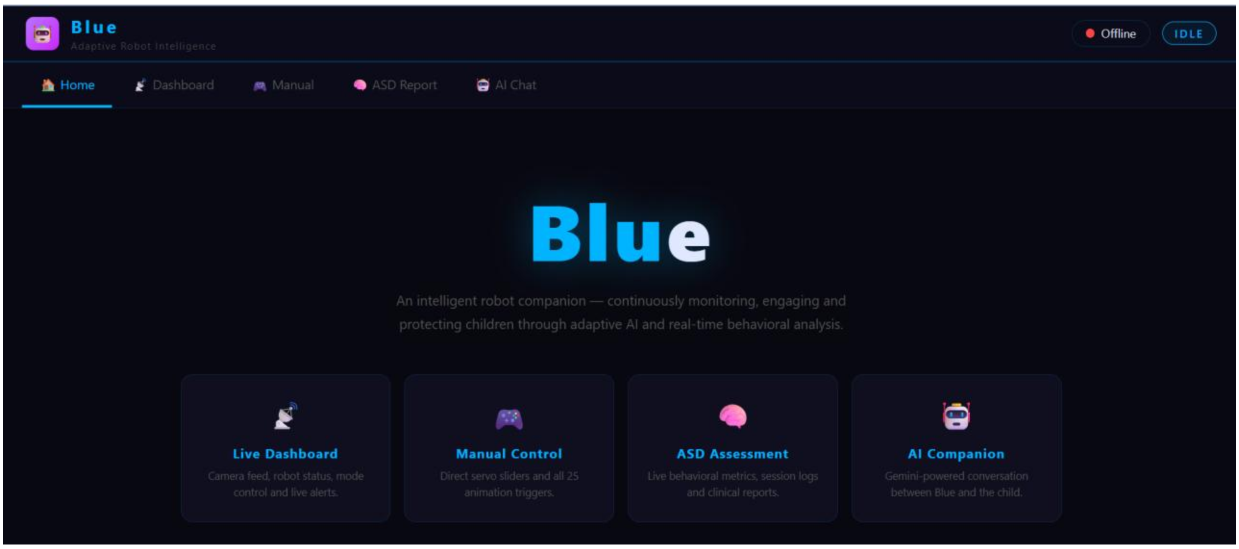

Fig 5.8 Web dashboard interface of the caregiver for providing live monitoring, manual control, ASD assessment, and AI interaction features.

CHAPTER 6

CONCLUSION

6.1 Review

In this report, the entire process of developing a smart robot assistant for children suffering from Autism Spectrum Disorder is discussed. In Chapter 1, the emphasis is on the challenges faced with ASD detection currently: It is not accessible, episodic, and subjective. The rationale for an always-on and deployable at home solution is given along with the goal of the proposed system which falls under SDGs 3 and 9. Chapter 2 discusses twenty-three papers on different areas related to the subject of ARIA including ASD detection using machine learning, pose estimation, recognition of self-injurious behavior, real-time detection using YOLO, and social robots for ASD intervention. The core idea behind how the working, and how combining hardware expressiveness and software intelligence makes sense were explained in Chapter 3. In Chapter 4, a comprehensive description of all the components making up the system was presented in terms of two interconnected layers. At the hardware layer, we have the Raspberry Pi 5, PCA9685 servo controller, ten servos, L298N motor controller, two DC motors, Lithium polymer batteries, and buck converters LM2596 forming four subsystems (power supply, computation, perception, and action). At the software layer, a multi-threaded Python back-end, YOLOv8, OpenPose library, Grok API conversational service, Firebase Realtime database, and a React-based application running on Vercel make up the solution.

6.2 Conclusion

1. An innovative design implementation process is discussed to design the whole ASD diagnosis system that will incorporate software as well as hardware components. This will enable continuous and intelligent monitoring of behavioral patterns of children.

2. The hardware captures reliable data capturing from cameras. Meanwhile, on the software front, multi-threading, real-time operations, and cloud connection will be used to ensure that the performance of the program is optimal and efficient enough. Different modes (monitoring and learning modes) will be available.

3. The system will be utilizing advanced algorithms such as pose estimation, facial emotion recognition, eye contact, and machine learning classification algorithms to better understand

the complexity of behavior analysis. Self-harm detection and adaptive learning will improve the performance even more.

4. Real-time self-injury detection software is provided based on YOLOv8n-pose, capable of identifying ten distinct classes of injurious behaviors from camera input and delivering audio intervention response immediately. The safety feature works round-the-clock without the requirement for caregiver supervision.

5. Behavioral analysis of children with ASD occurs 24/7 utilizing skeletal keypoint analysis via OpenPose and generates five clinically important measures such as body symmetry, posture score, engagement score, and bilateral wrist velocity resulting in structured session reports accessible to parents and clinicians via a web-based platform

6. Integration with Firebase and web interface will allow the system to store and visualize data instantly, while at the same time providing caregivers with remote access.

Together, these measures create an integrated approach that ensures consistent, evidence-based ASD services at home. This cuts down on unnecessary clinical appointments and provides valuable information about the child's behavior for both caregivers and clinicians.

6.3 Future Work

In terms of future work, incorporating wearable sensor information such as accelerometers, gyroscopes, and EMG could potentially enhance this system, particularly when video monitoring is not feasible due to privacy issues. By incorporating this information with existing vision-based features, detection accuracy could potentially increase, and false positives could potentially be minimized. This work contributes to the advancement of robust and non-intrusive assistive technologies for individuals with autism spectrum disorder.

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PUBLICATION DETAILS

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